

Table 1. Model parameters. (in the main text)

Table 2. OCV campaign settings (in main text)

Table 3. Impact of ORI with 75% vaccine coverage under the outbreak onset-triggered scenario. Values represent medians with interquartile ranges (IQRs).

Outbreaks	Week of ORI implementation	Outbreaks with ORIs implemented	% case reduction	Case averted	DALY averted	Case averted per 1,000 OCVs	DALY averted per 1,000 OCVs	Cost per case averted	Cost per DALY averted
All	1	1,192 (100%)	70% (49.8 - 82.5)	129 (41 - 375)	24 (0 - 128)	0.692 (0.188 - 2.399)	0.128 (0.001 - 0.679)	4,661.8 (984.8 - 17,916.7)	24,770.1 (4,065.8 - 4,679,419)
	5	1,192 (100%)	22.8% (3 - 48.4)	39 (3 - 193)	0 (0 - 46)	0.198 (0.015 - 1.093)	0.001 (0 - 0.238)	9,486.7 (1,843.7 - 40,532.9)	354,747.1 (8,338.3 - 24,678,219)
	9	1,084 (90.9%)	7.8% (0 - 29.5)	16 (0 - 112)	0 (0 - 23)	0.086 (0 - 0.658)	0 (0 - 0.122)	13,042.9 (2,574.7 - 60,025.6)	1,745,455 (11,816.2 - 40,406,869)
	13	822 (69%)	3.6% (0 - 20.7)	9 (0 - 87)	0 (0 - 22)	0.047 (0 - 0.491)	0 (0 - 0.067)	15,745.1 (2,912.8 - 75,909.7)	3,183,230 (13,251.2 - 51,572,816)
Top quartile attack rate	1	298 (25%)	79.1% (66.5 - 86.4)	518 (278 - 1,082)	107 (9 - 330)	4.747 (3.316 - 8.489)	0.849 (0.096 - 2.821)	415.3 (-15.8 - 872.9)	2,976.9 (-34.5 - 46,131)
	5	298 (25%)	45.4% (22.4 - 63.8)	282 (102 - 661)	44 (0 - 184)	2.798 (1.403 - 5.06)	0.354 (0.003 - 1.349)	959.7 (391.7 - 2,280.7)	8,557.4 (1,304.6 - 1,188,575)
	9	297 (24.9%)	18.9% (3.9 - 45.5)	133 (17 - 411)	14 (0 - 89)	1.302 (0.261 - 3.056)	0.048 (0 - 0.662)	1,734.9 (690.1 - 5,230.4)	20,433.3 (3,474.5 - 2,777,567)
	13	279 (23.4%)	8.3% (0.2 - 31.1)	56 (2 - 262)	0 (0 - 47)	0.546 (0.015 - 2.129)	0.001 (0 - 0.406)	2,915.4 (955.7 - 11,758.2)	47,963.7 (4,438.3 - 7,506,342)
Top quartile duration	1	298 (25%)	82% (72.9 - 88)	383 (155 - 895)	111 (24 - 294)	2.268 (0.77 - 4.948)	0.484 (0.111 - 1.624)	1,160.7 (357.5 - 4,485.8)	6,828.9 (1,104.4 - 33,519.1)

	5	298 (25%)	58.9% (45 - 71.2)	267 (98 - 661)	69 (19 - 214)	1.614 (0.501 - 3.803)	0.339 (0.043 - 1.097)	1,820.2 (591.9 - 7,149.3)	10,649.4 (2,201.6 - 89,189)
	9	298 (25%)	41.3% (25.6 - 57.4)	193 (60 - 492)	44 (0 - 154)	1.095 (0.3 - 2.786)	0.203 (0.003 - 0.768)	2,958.1 (941 - 12,656.2)	18,495.4 (3,672 - 1,275,264)
	13	298 (25%)	24.6% (12.9 - 44.1)	117 (34 - 329)	23 (0 - 109)	0.656 (0.159 - 1.95)	0.094 (0.001 - 0.513)	5,114.1 (1,459.4 - 24,635)	41,847.5 (6,339.6 - 6,694,783)
Top quartile size	1	298 (25%)	80% (68.3 - 86.9)	747 (496 - 1,446)	201 (46 - 444)	3.582 (1.578 - 7.418)	0.563 (0.111 - 1.999)	632.1 (85.9 - 1,805.2)	4,873.6 (322.8 - 27,860.5)
	5	298 (25%)	46.9% (25.1 - 65.7)	440 (215 - 954)	79 (1 - 236)	2.049 (0.567 - 4.336)	0.212 (0.004 - 1.006)	1,337.4 (475.1 - 5,359.3)	12,670 (1,966.5 - 482,108.3)
	9	296 (24.8%)	20.5% (3.7 - 47.4)	224 (42 - 587)	23 (0 - 138)	0.867 (0.123 - 2.57)	0.079 (0 - 0.501)	2,527.3 (808.5 - 11,251.5)	22,414.8 (4,107.8 - 2,650,052)
	13	271 (22.7%)	10.6% (0.5 - 35.4)	115 (7 - 409)	1 (0 - 106)	0.386 (0.019 - 1.817)	0.004 (0 - 0.402)	3,481.8 (1,090.1 - 19,323.1)	39,497.6 (4,083.6 - 6,598,464)

Table 4. Impact of ORI with 75% vaccine coverage under the case-triggered scenario. Values represent medians with interquartile ranges (IQRs).

Week of ORI implementation after the trigger	Case trigger thresholds (cumulative case)	Outbreaks with ORIs implemented	% case reduction	Case averted	DALY averted	Case averted per 1,000 OCVs	DALY averted per 1,000 OCVs	Cost per case averted	Cost per DALY averted
1	1	1,192 (100%)	69.8 (49.3 - 82.4)	129 (41 - 375)	24 (0 - 128)	0.691 (0.186 - 2.399)	0.127 (0.001 - 0.679)	4,645.5 (981.7 - 17,889.8)	24,797.8 (4,053.6 - 4,708,768)
1	10	1,179 (98.9%)	59.7 (33.4 - 76.5)	117 (29 - 361)	23 (0 - 113)	0.574 (0.126 - 2.254)	0.079 (0 - 0.592)	4,960.2 (1,004.5 - 20,384.9)	30,676.1 (4,532.6 - 7,235,077)
1	50	1,008 (84.6%)	43 (17.9 - 65.6)	111 (24 - 361)	21 (0 - 105)	0.543 (0.11 - 2.348)	0.049 (0 - 0.505)	5,269.6 (1,031.6 - 22,860)	38,239 (5,184.4 - 9,000,597)
1	140	716 (60.1%)	33.8 (13.2 - 57.2)	142 (34 - 443)	22 (0 - 115)	0.679 (0.13 - 2.631)	0.047 (0 - 0.507)	4,231.5 (922.2 - 20,119.2)	44,770 (5,250.1 - 6,439,389)
9	1	1,082 (90.8%)	7.8 (0 - 29.5)	16 (0 - 112)	0 (0 - 23)	0.086 (0 - 0.659)	0 (0 - 0.122)	13,073.7 (2,572.8 - 60,159.3)	1,730,823 (11,804.8 - 40,288,062)
9	10	1,013 (85%)	6.4 (0 - 27.2)	16 (0 - 116)	0 (0 - 23)	0.08 (0 - 0.677)	0 (0 - 0.116)	12,501.8 (2,564.2 - 61,947.2)	1,632,765 (12,023.6 - 42,082,255)
9	50	858 (72%)	4.3 (0 - 22.4)	14 (0 - 126)	0 (0 - 23)	0.066 (0 - 0.659)	0 (0 - 0.1)	11,614.7 (2,308.9 - 67,551.6)	1,362,625 (11,581.8 - 40,088,644)
9	140	613 (51.4%)	3.7 (0 - 18.4)	21 (0 - 148)	0 (0 - 24)	0.09 (0 - 0.744)	0 (0 - 0.107)	10,219.2 (1,998 - 48,208)	963,331.3 (10,357.1 - 29,151,154)